

Answers

Q	Ans	Q	Ans	Q	Ans
1	b	2	a	3	c
4	a	5	d	6	b
7	c	8	d	9	c
10	b	11	c	12	a
13	a	14	c	15	b
16	d	17	a	18	b
19	d	20	c	21	b
22	a	23	a	24	c
25	b	26	d	27	a
28	c	29	a	30	d
31	b	32	a	33	c
34	d	35	d	36	b
37	c	38	a	39	b
40	d	41	c	42	a
43	b	44	d	45	c
46	a	47	b	48	c
49	a	50	d	51	b
52	c	53	a	54	d
55	b	56	a	57	c
58	a	59	d	60	b

Selected working

- **Q1–3.** Agriculture = deliberate cultivation of crops + rearing of animals (text 1.1). Shifting cultivation abandons land after a few years; settled agriculture keeps the same plot continuously.
- **Q7–8.** Shifting cultivation cleared land for only one or two seasons; hunter-gatherers still live in desert areas, the Arctic region, and pockets of tropical rain forest.
- **Q18–19.** Vavilov’s concept of centers of origin; Ethiopia is a center for teff, coffee, enset and sorghum (standard knowledge, complementing the text).
- **Q27.** Industrialized-region eras: Mechanical (1930–1950), Chemical (1950–1970), Biotechnology/IT (1970–present).
- **Q28–31.** Ethiopia has about 51 million ha of arable land, only 20% cultivated; the five major crops (teff, wheat, maize, sorghum, barley) cover 75% of cultivated area; these grains supply 64% of calories.
- **Q38–39.** Monocots store food in the endosperm (starch); dicots like legumes and oil seeds store food in the cotyledons (protein, oil, carbohydrates).
- **Q45–46.** C3 crops: the first stable compound has 3 carbon atoms (wheat, teff, soybean); C4 crops accumulate CO₂ in the chloroplast and give high photosynthetic rates at low CO₂ (sorghum, maize, sugar cane).
- **Q48–56.** Mono cropping = one crop per plot (efficient but risk of total failure); mixed cropping = two or more crops at once; intercropping = more than one crop in regular rows at the same time; relay planting = second crop planted after the first has grown alone; rotation improves soil fertility especially with legumes.
- **Q59–60.** The Konso people practice traditional terracing and irrigation; highland farmers use kitchen ash to fertilize the soil and reduce acidity, and smoke to reduce frost damage.